PROJECT PROPOSAL

PII Filter

Masking personal information before it reaches the LLM

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Outline

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01

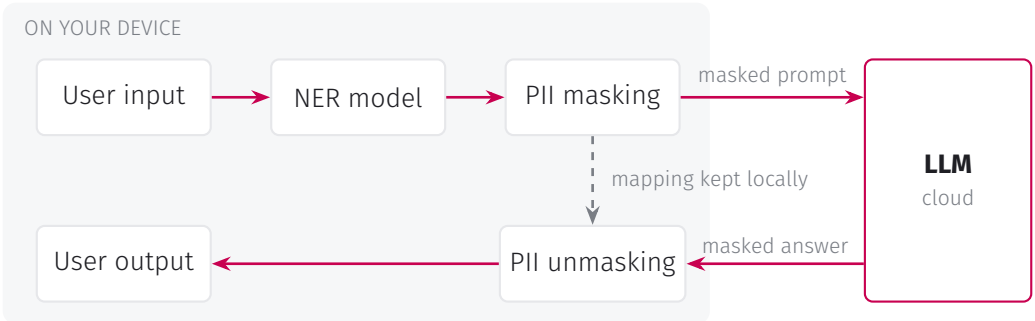
Problem Statement

A Concrete Example

YOU WRITE	Email Dr. Kim about my March 3 visit.
LLM RECEIVES	Email <NAME> about my <DATE> visit.
LLM ANSWERS	Dear <NAME>, about my <DATE> visit…
YOU READ	Dear Dr. Kim, about my March 3 visit…

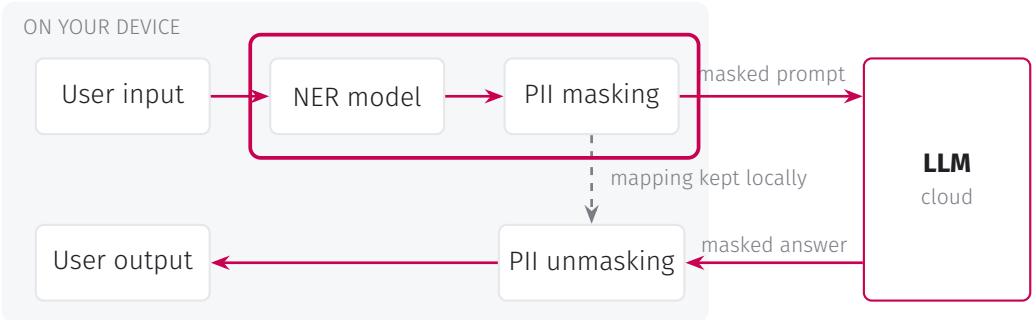
CLOUD

Proposed Pipeline



The LLM only ever sees placeholders, never the real personal information.

Proposed Pipeline



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Problem Statement

RQ1 Clean text

On clean text, how much better are fine-tuned BERT and GLiNER than rule-based systems across diverse datasets?



RQ2 Non-standard forms

How much does each system lose when PII is written in a non-standard form?

CLEAN	minji@example.com
NOISY	minji@exmaple.com

Rules: Microsoft Presidio (rules only) BERT: gravitee-io/bert-small-pii-detection GLiNER: urchade/gliner_multi_pii-v1

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Background

Named Entities and PII

Named Entity Recognition

Find the words that name something, and say **what kind of thing** it is.

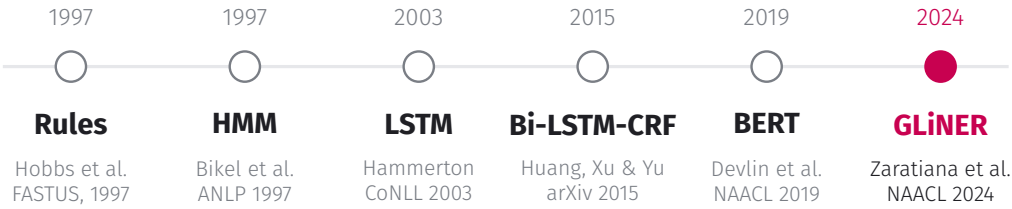
PII detection

The same task, applied to information that can identify a person.

Reach Louis NAME at louis@mail.com EMAIL or 010-1234-5678 PHONE.

Jurafsky & Martin, *Speech and Language Processing*

From Bi-LSTM-CRF to GLiNER



Each step needs less manual work and adapts better to new kinds of entities.

PII Masking for LLM Prompts

Tools

since 2023

Mask the PII, then put it back by simple replacement.

PrivateGPT LangChain LLM Guard

Research

2023 – 2025

Add a local language model to rewrite or restore the text.

Hide and Seek PAPILLON Rescriber

Private AI 2023 LangChain 2023 Protect AI 2023 Chen et al. 2023 Li et al. 2025 Zhou et al. 2025

03

Motivation

Who Is Concerned?

Companies

Trade secrets

Employees paste code, client data and internal documents into chatbots.

Hospitals

Patient records

Medical data is highly sensitive and strictly protected by law.

Individuals

Private life

People share names, addresses and health details, often without noticing.

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Metric and Benchmark

Benchmarks

Entities evaluated in this benchmark

NAME	EMAIL	PHONE
ADDRESS	DATE_OF_BIRTH	CARD_NUMBER

Datasets used in this benchmark

ai4privacy 500k
Short sentences

Nemotron-PII
Forms, emails, notes

Gretel PII finance
Financial documents

Ai4Privacy 2025 Steier et al. 2025 Watson et al. 2024

Metrics

A missed PII leaks. An extra mask only costs some context.

Precision

$$P = \frac{TP}{TP + FP}$$

Recall

$$R = \frac{TP}{TP + FN}$$

F1

$$F1 = \frac{2\,P\,R}{P + R}$$

MAIN SCORE

F2

$$F2 = \frac{5\,P\,R}{4\,P + R}$$

05

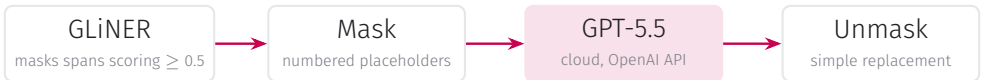
Experiment Design

Experiment Design



Every test sentence is scored twice: as it is, and with noise added.

Toy Experiment: Setup



CLEAN Write a short message to my landlord. I'm Minji Park from 41 Maple Street, and my heater is broken. He can call me at 010-2345-6789.

NOISY write a short message to my landlord im minji park from 41 maple stret and my heater is broken he can call me at 010 2345 6789

Noise: lowercase, no punctuation, one typo, spaces in the phone number.

Toy Experiment: Example 1 (Clean)

LLM RECEIVES	Write a short message to my landlord. I'm <NAME_1> from <ADDRESS_1>, and my heater is broken. He can call me at <PHONE_1>.
LLM ANSWERS	Hi, this is <NAME_1> from <ADDRESS_1>. My heater is broken and needs repair. Please call me at <PHONE_1> when you can. Thank you.

CLOUD

YOU READ Hi, this is Minji Park from 41 Maple Street. My heater is broken and needs repair. Please call me at 010-2345-6789 when you can. Thank you.

Answers are shown in full, as returned by gpt-5.5.

Toy Experiment: Results

The loop works on both prompts, in all 3 runs.

PRIVATE?	**Yes**	no real value was sent
ENOUGH CONTEXT?	**Yes**	all 3 placeholders were used
USABLE ANSWER?	**Yes**	all 3 values were restored

The noise comes back: the noisy answer reads “this is minji park from 41 maple stret”.

Confidence drops: the lowercase name falls from 0.999 to 0.686, close to the 0.5 cut-off.

Scores: urchade/gliner_multi_pii-v1 (results/toy/openai/toy_results.md).

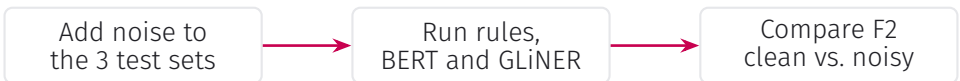
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Conclusion

Summary and Next Steps

THE IDEA	Mask the PII on the device: the LLM never sees the real values.
THE QUESTION	Rules, BERT or GLiNER: which finds PII best, which resists noise?
FIRST SIGNAL	The toy loop works, but noise lowers the detector's confidence.

Next steps



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Handling PII data in LangChain, 2023.
PresidioReversibleAnonymizer: anonymize the prompt, de-anonymize the answer with a mapping.

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Thank you

Questions?